Hyejin Kim

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SUMMARY

Ph.D. candidate in physics developing physics-informed machine-learning methods for quantum information science. Research spans measurement-driven characterization of quantum many-body dynamics, scalable quantum error-correction decoding and hardware-aware compilation.

Research interests: Quantum information and dynamics; machine learning for quantum systems; quantum error correction; hardware–software co-design; interpretable machine learning for physics and computation.

EDUCATION

Aug. 2022 – exp. Aug. 2027 — **Cornell University** — *NY, USA*
PhD candidate in Physics
M.S. in Physics, May 2025
- Advisor: Prof. Eun-Ah Kim

2018 – 2022 — **Gwangju Institute of Science & Technology (GIST)** — *Gwangju, S. Korea*
B.S. in Physics, minor in Mathematics (Major GPA: 4.47/4.5)
- Thesis: Variational Autoencoder for the metal-insulator transition in DMFT
- Advisor: Prof. Dong-Hee Kim

RESEARCH EXPERIENCE

2022 – pres. — **Kim Research Group, Cornell Physics** — *NY, USA*
Graduate Research Assistant (Advisor: Prof. Eun-Ah Kim)

2025 – 2026
- **Reinforcement learning agent for Hardware-aware reconfigurable neutral Atom Array Compilation**
  - ··· Developed Neutral Atom Compiler Agent (NACA): RL compiler for neutral atom quantum processors whose training objective is the device's own fidelity functional evaluated on the executed transport trajectory.
  - ··· NACA exceeds state-of-the-art heuristic compiler up to 118 atoms and transfers zero-shot to unseen circuit families.

- **Multi-Core sparse-attention Circuit Decoder for Surface code**
  - ··· Currently working on an attention-based circuit-level surface code decoder designed to scale across code distances and logical circuit structure.

2024 – 2026
- **Learning the phases of monitored quantum dynamics**
  - ··· Developed a scalable, data-centric framework for detecting measurement-induced phase transitions (MIPTs) in monitored circuits without post-selection or classical simulation, enabling experimental observability on near-term quantum hardware
  - ··· Manuscript under review in PRX Intelligence. Experimental collaboration with Quantinuum in the ongoing follow-up work.

2022 – 2024

- **Attention to Quantum Complexity**
  - ··· Introduced Quantum Attention Network (QuAN) that leverages the power of attention mechanisms driving large language models (LLMs) to characterize the complexity of quantum states from bit string data.
  - ··· Resolved three open problems: (i) witnessing entanglement scaling transition, (ii) detecting the growth of quantum complexity in deep random quantum circuits despite the noise, (iii) revealing mixed-state topological order of toric code.
  - ··· Collaboration with Google Quantum AI and MIT. Published in Science Advances.

2020 – 2022 **Computational Many-body Physics Group, GIST** *Gwangju, S. Korea*

Undergraduate Research Intern (Advisor: Prof. Dong-Hee Kim)

- Variational Autoencoder for the metal-insulator transition in DMFT
- Interpretation of a minimal neural network to learn the metal-insulator transition
- Machine-learning accelerated Monte-Carlo algorithm for 2D spin model

PUBLICATIONS

[1] **H. Kim**, A. Kumar, Y. Zhou, Y. Xu, R. Vasseur, E.-A. Kim, Learning measurement-induced phase transitions using attention, *arXiv:2508.15895* (under review in PRX Intelligence, 2026).

[2] J. Tahmassebpur, A. Bhuiyan, **H. Kim**, O. Lesser, Neural network robustness under heavy input corruption, *Phys. Rev. Research* (2026), doi:10.1103/qkrg-wbbj.

[3] **H. Kim**, Y. Zhou, Y. Xu, K. Varma, A. H. Karamlou, I. T. Rosen, J. C. Hoke, C. Wan, J. P. Zhou, W. D. Oliver, Y. D. Lensky, K. Q. Weinberger, E.-A. Kim, Attention to Quantum Complexity, *Sci. Adv. 11*, eadu0059 (2025).

[4] **H. Kim**, D. Kim, D.-H. Kim, Minimal Neural Network to Learn the Metal-insulator Transition in the Dynamical Mean-field Theory, *NPSM 2022,* **72**, 487-494 (2022).

[5] **H. Kim**, Y. Xu, J. P. Zhou, E.-A. Kim, Reinforcement learning for Hardware-Aware Reconfigurable Atom Array Compilation (in preparation, 2026)

PRESENTATIONS

Invited talks

[1] **H. Kim**. Learning Atom Trajectories: Hardware-aware Compilation for Reconfigurable Neutral-Atom Arrays, *Invited Talk, Marquardt Division, Max Planck Institute for the Science of Light*, Erlangen, Germany, Oct. 2026.

[2] **H. Kim**. AI for Quantum: Decoding logical circuit and Compilation on Neutral atom platform, *Research Seminar, LG Electronics*, Seoul, S. Korea, Jun. 2026.

[3] **H. Kim**. Reinforcement learning for hardware-aware reconfigurable atom array compilation, *Lunch Seminar, Department of Quantum Information Engineering, Sungkyunkwan University*, Suwon, S. Korea, Jun. 2026.

[4] **H. Kim**. Attention to Quantum Complexity: Learning measurement-induced phase transitions, *Group Seminar, Department of Physics, KAIST*, Daejeon, S. Korea, Jun. 2026.

[5] **H. Kim**. Learning Measurement-Induced Phase Transition Using Attention, *QST Seminar, Seoul National University*, Seoul, S. Korea, Jun. 2026.

[6] **H. Kim**. Attention to Quantum Complexity: Learning measurement-induced phase transitions, *Department of Physics, Sejong University*, Seoul, S. Korea, Jun. 2026.

[7] **H. Kim**. Learning Measurement-Induced Phase Transition Using Attention, *Seminar, Korea Institute for Advanced Study*, Seoul, S. Korea, Jun. 2026.

[8] **H. Kim**. Reinforcement Learning for Hardware-Aware Reconfigurable Atom Array Compilation, *AI and Natural Sciences Seminar, Korea Institute for Advanced Study*, Seoul, S. Korea, Jun. 2026.

[9] **H. Kim**. Learning Measurement-Induced Phase Transition Using Attention, *Special Seminar, GIST*, Gwangju, S. Korea, Jun. 2026.

[10] **H. Kim**. Learning Measurement-Induced Phase Transition Using Attention, *Seminar, Yonsei University*, Incheon, S. Korea, Jun. 2026.

[11] **H. Kim**, Attention to quantum complexity: learning phases of monitored quantum dynamics, In *APS Joint March and April Meeting Global Physics Summit*, 2025

Contributed talks

[12] **H. Kim**, Learning Hardware-Aware Compilation for Reconfigurable Neutral-Atom Arrays, Lightning talk, AI for Decoding/Logical Circuit Compilation session, In *AI for Quantum Control and Measurement Workshop*, Berkeley Quantum Nexus, Berkeley, CA, Oct. 2026

[13] **H. Kim**, Y. Xu, J. P. Zhou, K. Q. Weinberger, E.-A. Kim. Reinforcement Learning for Hardware-Aware Reconfigurable Atom Array Compilation, In *APS Global Physics Summit*, 2026

[14] **H. Kim**, A. Kumar, Y. Zhou, Y. Xu, R. Vasseur, E.-A. Kim. Learning the phases of monitored quantum dynamics - II, In *APS Joint March and April Meeting Global Physics Summit*, 2025

[15] **H. Kim**, Y. Zhou, Y. Xu, C. Wan, J. P. Zhou, Y. Lensky, J. Hoke, P. Roushan, K. Weinberger, E.-A. Kim. Attention to complexity III: learning the complexity of random quantum circuit states, In *American Physical Society March Meeting*, 2024

[16] **H. Kim**, K. Varma, C. Wan, Y. Zhou, Y. Lensky, K. Weinberger, E.-A. Kim. Detecting quantum complexity using transformer-based neural network (II), In *American Physical Society March Meeting*, 2023

[17] **H. Kim**. Attention to Quantum Complexity, In *Cornell Quantum Day Fall 2023*

[18] **H. Kim**, D. Kim, D.-H. Kim. Interpretation of a minimal neural network to learn the metal-insulator transition in the dynamical mean-field theory, In *2021 Korean Physical Society Conference in Gwangju-Chonnam Branch*, 2021

SKILLS

Programming Languages

- Language: Python, C/C++
- Machine learning framework: PyTorch, Gymnasium
- Quantum simulation: Qiskit, Cirq, Pytket, Guppy, Stim

ACADEMIC HONORS

2020 – 2021 **National Scholarship for Science & Engineering**, Korea Student Aid Foundation, full fund

2018 – 2019 **Korean Government Scholarship**, GIST

2018 – 2020 **Academic Scholarship to excellence**, GIST

TEACHING EXPERIENCE

Cornell University *NY, USA*

- Teaching assistant
  - *Fall 2025* · · · PHYS 2214: Oscillations, Waves, and Quantum Physics: Manage discussion sections, lab sections, homework, and exams.
  - *Fall 2022* · · · PHYS 2213: Electromagnetism: Manage discussion sections, homework, and exams.